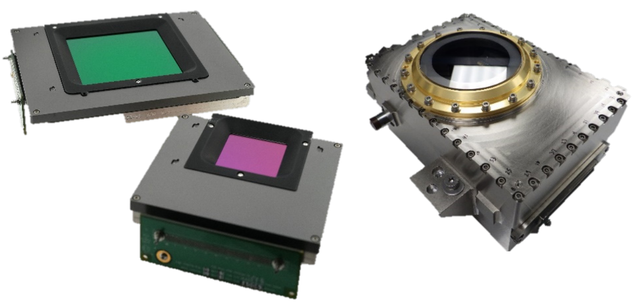
GeoSnap-10 High Performance Focal Plane Array (FPA)

Designed for the most demanding Earth observation, geospatial, and missile warning, tracking and defense applications.

The GeoSnap-10 FPA leverages decades of advanced infrared design from Teledyne Space Imaging to provide market leading detector array size and sensitivity at high production rates.

FEATURE	BENEFIT
Proprietary FPA hybridization process integrating large format HgCdTe detectors with high performance GeoSnap digital ROICs	• Stitchable detector blocks of 2K×1K pixels (common sizes include 4K×4K, 6K×6K, 8K×8K; others possible) • Largest format IR focal planes on-orbit, delivering critical data today
Proven HgCdTe detector	• Detection from visible through LWIR • Configurable cutoff wavelengths from SWIR to LWIR • High Quantum Efficiency and low Dark Current
Digital ROIC architecture	• 14-bit ADC on-chip digital conversion • Dual gain settings with low read noise • Anti-blooming
Established production lines and material supply chain	• Demonstrated capability supporting large quantity orders required for proliferated space architectures (High MRL)
Radiation hardened	• Fully radiation hardened for space environments • TRL-9 for LEO
Laboratory Focal Plane Electronics (FPE)	• FPE available with EDU or ROIC-only modules for quick and easy lab integration of FPMs • Enables SW/FW integration prior to flight deliveries



PACKAGING OPTIONS

- Standard packaging includes the FPA packaged in a connectorized rugged packaging and the associated interface cable(s)
- Options are available for packaging in a complete Integrated Dewar Assembly (IDA) upon request

GeoSnap-10 4K×4K and 8K×8K FPMs are currently enabling critical applications requiring state of the art large format detectors that are production ready.

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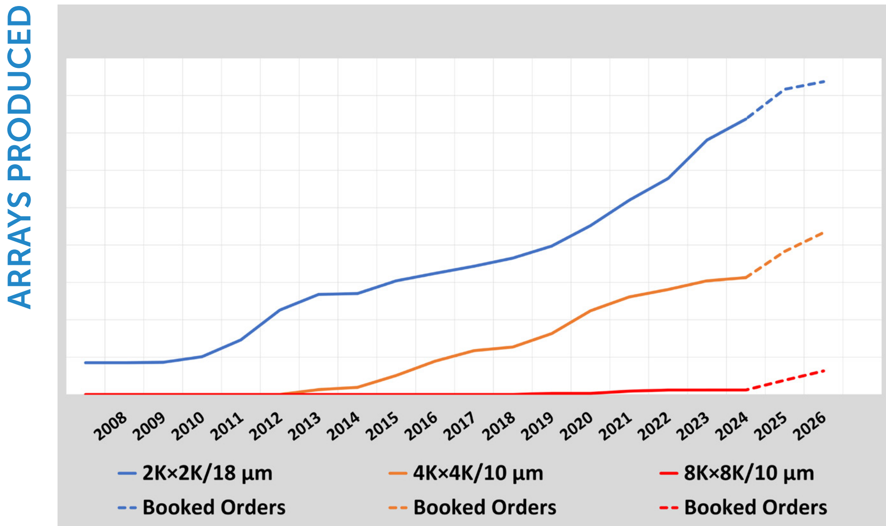
General Focal Plane Module (FPM) properties are provided in the following table. Detailed performance specifications, product ICDs, and manuals are available upon request.

PARAMETER	UNITS	GEOSNAP-10 STANDARD VALUES
Cutoff wavelength (λco)	µm	Visible and infrared (VNIR [1.05 µm] through VLWIR [14 µm]) Operation as low as 400 nm available with substrate thinning
Pixel architechture		10 µm unit pixel, Capacitive transimpedance amplifier (CTIA)
Operating temperature	K	45 to 300 (80 – 140 typical)
Readout mode		Snapshot, Integrate While Read (IWR)
Digital conversion and output		14-bit ADC on-chip - Current mode logic (CML)
Serial bit data		8b/10b encoded
Output data bit rate / port	Gbps	1.6 to 3.2
Number of ports		8K = 32 ports, 6K = 24 ports, 4K = 16 ports
Required input voltages	V	0.4, 1.8, 3.3, 3.6
Focal Plane Module size	inch	4K×4K: 4.5" x 4.6" x 0.85" 8K×8K: 6.5" x 7.0" x 1.5"
Focal Plane Module mass	kg	4K×4K: <0.6kg 8K×8K: <2.2kg

* New High Dynamic Range (HDR) ROIC option planned for 2025



Teledyne Space Imaging has an active production line for GeoSnap products with a demonstrated ability to consistently deliver large format focal planes at high quantity.

Greater than 1,000 GeoSnap arrays have been produced to date, with over 100 being delivered to the SDA's Tracking Layer program.

Approved for public release: Teledyne Approved.
Information subject to change – values typical unless stated.

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